

Assignment 5

Assignment Title

Design and Develop a Dynamic Web Application using Node.js, Express, React, and MongoDB (College Website + Event Attendance System)

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Objectives

- To develop a full-stack web application using Node.js and Express
 - To understand backend API development and routing
 - To integrate MongoDB for data storage
 - To build an interactive frontend using React
 - To design a real-world system for event attendance and college website
 - To implement CRUD operations using REST APIs
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Outcomes

- Learned full-stack development (Frontend + Backend)
 - Understood API creation and integration
 - Implemented MongoDB database operations
 - Built a responsive and interactive UI using React
 - Gained experience in real-world application development
 - Improved problem-solving and debugging skills
-

Theory Concept in Brief

A full-stack web application consists of:

1. Frontend

The frontend is the user interface where users interact with the system.

In this project, React is used to create a dynamic UI with components and state management.

2. Backend

The backend is responsible for handling requests, processing data, and communicating with the database.

Node.js and Express are used to create REST APIs.

3. Database

MongoDB is a NoSQL database used to store data in JSON format.

Mongoose is used to define schemas and interact with MongoDB.

Technologies Used

Frontend Technologies

- HTML, CSS → UI Design
- JavaScript → Logic
- React.js → Dynamic UI

Backend Technologies

- Node.js → Runtime environment
- Express.js → Web framework

Database

- MongoDB → NoSQL database
- Mongoose → ODM (Object Data Modeling)

Core Functionalities Implemented

ASS5_1: College Website

- Dynamic college homepage
- Navigation (Home, Programs, Faculty, Admissions, Contact)
- Admission form handling
- Contact form submission
- Data stored in MongoDB
- API integration for fetching applications

As seen in your backend: routes like /apply, /contact, /api/applications are implemented

ASS5_2: Event Attendance System

- Event registration form
- Form validation
- Data submission using Fetch API
- Storage in MongoDB database
- Display registered events dynamically

React frontend handles state and API calls
Express backend stores and retrieves data

Methodology / Steps Followed

Step 1: Designed UI using HTML and CSS

Step 2: Created React components for event form

Step 3: Implemented state management using React Hooks

Step 4: Developed backend using Node.js and Express

Step 5: Created REST APIs for data handling

Step 6: Connected MongoDB using Mongoose

Step 7: Implemented form submission and data storage

Step 8: Displayed stored data dynamically

Step 9: Tested application and handled errors

Program Code (Overview)

Frontend (React)

The event attendance form collects user data and sends it to backend:

```
fetch("http://localhost:5000/attend-event", {  
  method: "POST",  
  headers: {  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify({  
    name, email, phone, college, event, date  
  })  
});
```

Backend (Express + MongoDB)

```
app.post("/attend-event", handleAttendance);  
app.get("/get-events", getEvents);
```

The backend stores data in MongoDB using schema:

```
const eventSchema = new mongoose.Schema({  
  name:String,  
  email:String,  
  phone:String,  
  college:String,  
  event:String,  
  date:String  
});
```

Real-Life Application

- College admission systems
 - Event management platforms
 - Online registration systems
 - Student data management systems
 - Web-based administrative systems
-

Output Screenshots

Include:

- College Website Homepage
- Event Attendance Form
- MongoDB Data Output
- Registered Events Display

Event Attendance

FULL NAME

Enter your full name

EMAIL ADDRESS

your.email@example.com

PHONE NUMBER

Enter your phone number

COLLEGE NAME

Enter your college name

EVENT NAME

Which event are you attending?

EVENT DATE

dd-mm-yyyy



MARK ATTENDANCE

MongoDB Compass - TY_FSD/EventDB.events

Connections Edit View Collection Help

Compass

My Queries

Data Modelling

CONNECTIONS (3)

Search connections

PCOOE

- TY_FSDL
 - ASS5_2
 - Pcooe
 - cdmin
 - college_website
 - config
 - eventDB
 - events
 - local
 - loginDB
 - pcooe
 - practice
 - test
 - timetableDB
- Pcooe

TY_FSDL > eventDB > events

Documents 7 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE

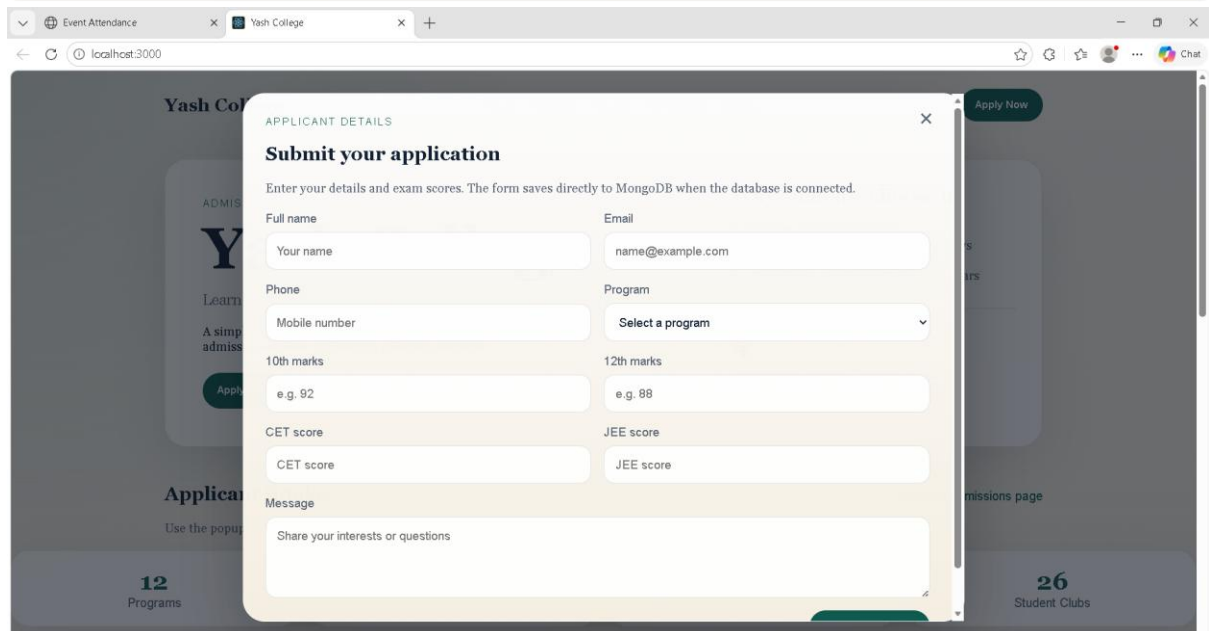
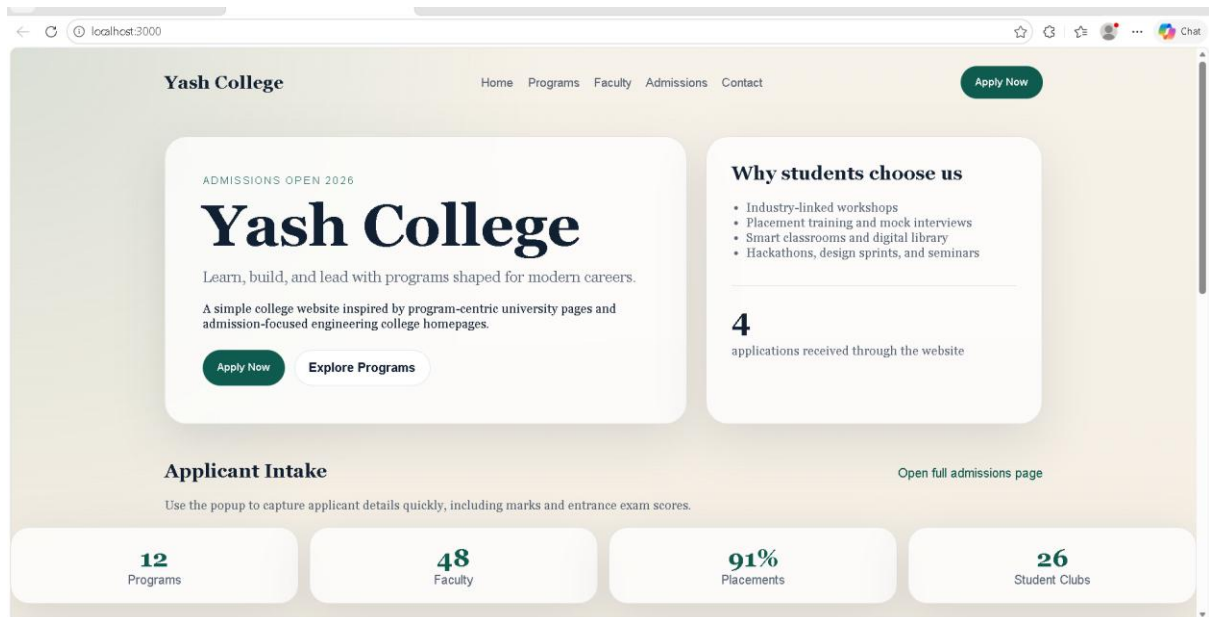
25 1-7 of 7

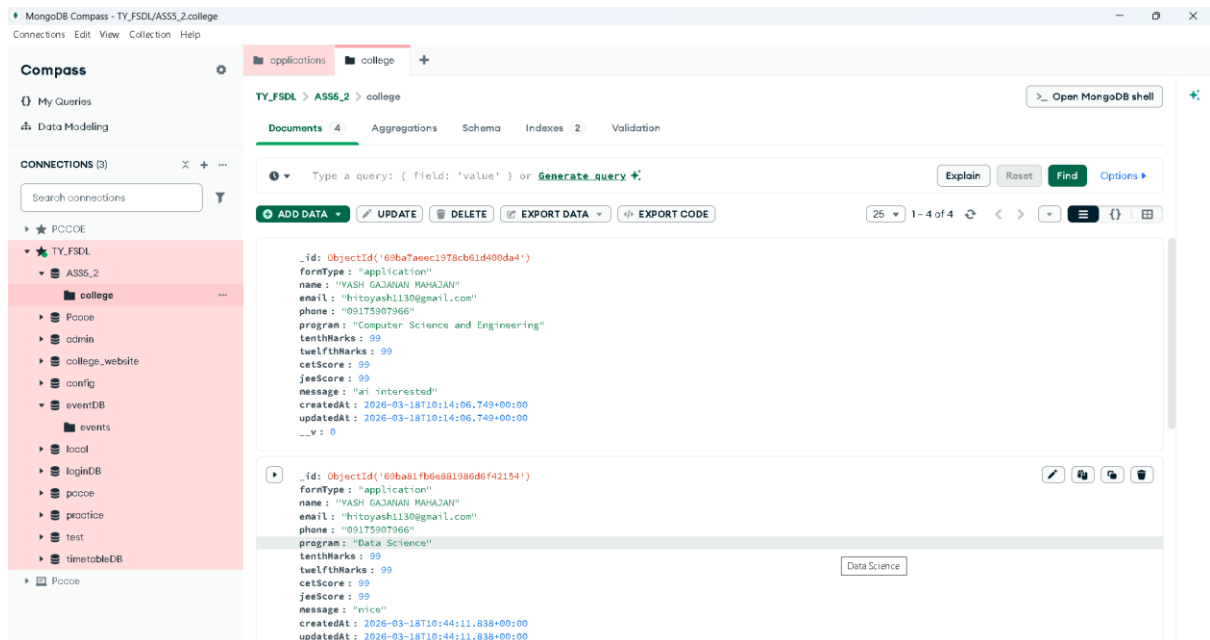
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  "name": "YASH GAJANAN MAHAJAN",  
  "email": "hitoyash1130@gmail.com",  
  "phone": "9175907966",  
  "college": "PCCOE",  
  "event": "hackathon",  
  "__v": 0  
}
```

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{ "_id": ObjectId("69b0ef4fcc65c2fabdfa546"),  
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  "phone": "9175907966",  
  "college": "PCCOE",  
  "event": "dy hackathon",  
  "__v": 0  
}
```

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{ "_id": ObjectId("69b0f8beccc65c2fabdfa548"),  
  "name": "Aryan",  
  "email": "yash.mahajan23@pccopune.org",  
  "phone": "9175907966",  
  "college": "PCCOE",  
  "event": "hackathin in pcooe",  
  "__v": 0  
}
```

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{ "_id": ObjectId("69ba1ebdf8be85eb9521ea2"),  
  "name": "prathamesh",  
  "__v": 0  
}
```





Link to codes-

<https://drive.google.com/drive/u/0/folders/1fXHbwqTHYSAZlr3XpMKJCilIFRvQRISf>

Conclusion

The assignment successfully demonstrates the development of a full-stack web application using modern technologies like Node.js, Express, React, and MongoDB. It helped in understanding frontend-backend integration, database management, and real-world application development.

The project enhances practical knowledge of web development and prepares for industry-level applications.